

14-mavzu: n-tartibli bir jinsli bo'lmagan o'zgarmas koeffitsiyentli chiziqli differensial tenglamalar.

Reja:

1. Noma'lum koeffitsiyentlar metodi.
2. Superpozitsiya prinsipi.

Faraz qilaylik, $P(x)$ ko'phad

$$P(x) = b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0, \quad b_m \neq 0, \quad (1)$$

ko'rinishda bo'lsin

Teorema. Agar (1) tenglamadagi σ son uchun $L(\sigma) \neq 0$ bo'lsa, u holda

$$L[y] = (b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0) e^{\sigma x}$$

(1) tenglama

$$y = (r_m x^m + r_{m-1} x^{m-1} + \dots + r_1 x + r_0) e^{\sigma x} \quad (2)$$

ko'rinishdagi xususiy yechimga ega.

$P(x)$ ko'phadning darajasi m bo'yicha matematik induksiya metodini qo'llaymiz. Dastlab $\deg P(x) = 0$ deymiz va $L[y] = b_0 e^{\sigma x}$ tenglama $y = r_0 e^{\sigma x}$ ko'rinishdagi xususiy yechimga egaligini ko'rsatamiz. (2) formulaga ko'ra

$$L[r_0 e^{\sigma x}] = r_0 e^{\sigma x} L(\sigma) = b_0 e^{\sigma x}.$$

Bundan $L[y] = b_0 e^{\sigma x}$ tenglama qanoatlanishi uchun $r_0 = \frac{b_0}{L(\sigma)}$ deyish kifoya ekanligi kelib chiqadi. Demak, $m = 0$ holida teorema isbotlandi. Endi induksiya farazini qilib, ya'ni $L[y] = \tilde{P}(x) e^{\sigma x}$, $\deg P(x) \leq m-1$, tenglamaning $y = \tilde{Q}(x) e^{\sigma x}$, $\deg Q(x) \leq m-1$,

ko'rinishdagi yechimga ega ekanligini ma'lum deb, $L[y] = P(x) e^{\sigma x}$, $\deg P(x) = m$, tenglamaning $y = Q(x) e^{\sigma x}$, $\deg Q(x) = m$, ko'rinishdagi yechimga egaligini isbotlashimiz kerak. Berilgan

$$L[y] = P(x) e^{\sigma x}, \quad \deg P(x) = m, \quad P(x) = (b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0) e^{\sigma x},$$

tenglamaning yechimini

$$y = r_m x^m e^{\sigma x} + u \quad (3)$$

ko'rinishda izlaymiz; bu yerda r_m hozircha noma'lum son, u yangi noma'lum funksiya. Demak,

$$L[r_m x^m e^{\sigma x} + u] = P(x) e^{\sigma x}, \quad \text{ya'ni}$$

$$r_m L[x^m e^{\sigma x}] + L[u] = (b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0) e^{\sigma x} \quad (4)$$

tenglik qanoatlanishi kerak. Endi r_m ni tanlash evaziga (4) tenglamadan x^m qatnashgan hadni yo'qotishga harakat qilamiz. (3) ayniyatdan

$$\begin{aligned} L[x^m e^{\sigma x}] &= (L(\sigma) x^m + C_m^1 L'(\sigma) x^{m-1} + \\ &+ C_m^2 L''(\sigma) x^{m-2} + \dots + L^{(m)}(\sigma)) e^{\sigma x} = \\ &= L(\sigma) x^m e^{\sigma x} + R(x) e^{\sigma x}; \end{aligned} \quad (5)$$

bu yerdagi $R(x) = C_m^1 L'(\sigma)x^{m-1} + C_m^2 L''(\sigma)x^{m-2} + \dots + L^{(m)}(\sigma)$ ko'phadning darajasi $\deg R(x) \leq m-1$. Induksiya faraziga ko'ra bu tenglama $u = \tilde{Q}(x)e^{\sigma x}$, $\deg Q(x) \leq m-1$, ko'rinishdagi yechimga ega. Demak, (1) formulaga ko'ra $L[y] = P(x)e^{\sigma x}$, $\deg P(x) = m$, tenglama $y = r_m x^m e^{\sigma x} + \tilde{Q}(x)e^{\sigma x}$, ya'ni $y = Q(x)e^{\sigma x}$, $Q(x) \stackrel{\text{def}}{=} r_m x^m + \tilde{Q}(x)$, $\deg Q(x) = m$, ko'rinishdagi yechimga ega. $\square$

Teorema. Agar (1) tenglamadagi σ son $L(\lambda) = 0$ xarakteristik tenglamaning k karrali ildizi bo'lsa, u holda

$$L[y] = (b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0) e^{\sigma x}$$

(1) tenglama

$$y = x^k (r_m x^m + r_{m-1} x^{m-1} + \dots + r_1 x + r_0) e^{\sigma x} \quad (6)$$

ko'rinishdagi yechimga ega.

$\square$ Yuqoridagi teoremaning isbotiga o'xshash fikr yuritamiz. $\deg P(x) = m$ bo'yicha matematik induksiya metodini qo'llaymiz, faqat endi sal nozikroq fikrlash kerak bo'ladi. Dastlab teoremaning shartiga ko'ra $L(\lambda) = (\lambda - \sigma)^k M(\lambda)$, $M(\sigma) \neq 0$ ($k \in \mathbf{N}$), ekanligini e'tirof etaylik. Demak,

$$L(\sigma) = L'(\sigma) = \dots = L^{(k-1)}(\sigma) = 0, L^{(k)}(\sigma) = k! M(\sigma) \neq 0. \quad (7)$$

$m=0$ bo'lsin. Ushbu

$$L[y] = b_0 e^{\sigma x} \quad (8)$$

tenglama $y = r_0 x^k e^{\sigma x}$ ko'rinishdagi yechimga ega bo'lishini ko'rsatishimiz kerak. Buning uchun $L[r_0 x^k e^{\sigma x}]$ ni hisoblaymiz. (7) formuladan (8) ga ko'ra quyidagini topamiz:

$$\begin{aligned} L[r_0 x^k e^{\sigma x}] &= r_0 (L(\sigma) x^k + C_k^1 L'(\sigma) x^{k-1} + \\ &\quad + C_k^2 L''(\sigma) x^{k-2} + \dots + L^{(k)}(\sigma)) e^{\sigma x} = \\ &= r_0 k! M(\sigma) e^{\sigma x}. \end{aligned}$$

Demak, agar $r_0 = \frac{b_0}{k! M(\sigma)}$ desak, u holda $y = r_0 x^k e^{\sigma x}$ funksiya (6) tenglamaning yechimi bo'ladi. Teorema $m=0$ holida isbot bo'ldi.

Endi induksiya farazini qilamiz, ya'ni $L[y] = \tilde{P}(x)e^{\sigma x}$, $\deg P(x) \leq m-1$, tenglama $y = x^k \tilde{Q}(x)e^{\sigma x}$, $\deg Q(x) \leq m-1$, ko'rinishdagi yechimga ega ekanligini ma'lum deymiz va $L[y] = P(x)e^{\sigma x}$, $\deg P(x) = m$, tenglamaning $y = x^k Q(x)e^{\sigma x}$, $\deg Q(x) = m$, ko'rinishdagi yechimga ekanligini isbotlaymiz. Berilgan $L[y] = P(x)e^{\sigma x}$, $P(x) = b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0$, $\deg P(x) = m$, tenglamaning yechimini

$$y = r_m x^{k+m} e^{\sigma x} + u \quad (9)$$

korinishda izlaymiz. Bu yerda $u = u(x)$ - yangi noma'lum funksiya, r_m - noma'lum son. Biz ularni berilgan tenglamaning qanoatlanishi, ya'ni

$$L[r_m x^{k+m} e^{\sigma x} + u] = P(x) e^{\sigma x}$$

shartidan topishimiz kerak. Bu tenglamani

$$r_m L[x^{k+m} e^{\sigma x}] + L[u] = (b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0) e^{\sigma x} \quad (10)$$

ko'rinishda yozib olamiz va r_m ni shunday tanlaymizki, natijada (9) tenglamada x^m qatnashmasin. Buning uchun $L[x^{k+m} e^{\sigma x}]$ ni hisoblaymiz. (10) formuladan (8) munosabatlarni hisobga olib topamiz:

$$\begin{aligned} L[x^{k+m} e^{\sigma x}] &= (L(\sigma) x^{k+m} + C_{k+m}^1 L'(\sigma) x^{k+m-1} + \dots + C_{k+m}^{k-1} L^{(k-1)}(\sigma) x^{m-1} + \\ &\quad + C_{k+m}^k L^{(k)}(\sigma) x^m + C_{k+m}^{k+1} L^{(k+1)}(\sigma) x^{m-1} + \dots + L^{(k+m)}(\sigma)) e^{\sigma x} = \\ &= (C_{k+m}^k L^{(k)}(\sigma) x^m + C_{k+m}^{k+1} L^{(k+1)}(\sigma) x^{m-1} + \dots + L^{(k+m)}(\sigma)) e^{\sigma x} = \\ &= (C_{k+m}^k k! M(\sigma) x^m + S(x)) e^{\sigma x}, \end{aligned} \quad (11)$$

bunda $S(x) = C_{k+m}^{k+1} L^{(k+1)}(\sigma) x^{m-1} + \dots + L^{(k+m)}(\sigma)$ ko'phadning darajasi $\deg S(x) \leq m-1$.

Endi (11) ni (10) ga qo'yib, $r_m = \frac{b_m}{C_{k+m}^k k! M(\sigma)}$ deb tanlab, u noma'lum funksiya uchun ushbu

$$L[u] = \tilde{P}(x) e^{\sigma x},$$

$$\tilde{P}(x) \stackrel{\text{def}}{=} (b_{m-1} x^{m-1} + \dots + b_1 x + b_0 - S(x)), \deg P(x) \leq m-1, \quad (12)$$

tenglamani hosil qilamiz. Induksiya faraziga ko'ra (12) tenglama $u = x^k \tilde{Q}(x) e^{\sigma x}$, $\deg Q(x) \leq m-1$, ko'rinishdagi yechimga ega. (11) formulaga ko'ra $L[y] = P(x) e^{\sigma x}$, $\deg P(x) = m$, tenglama esa

$$\begin{aligned} y &= r_m x^{k+m} e^{\sigma x} + u = r_m x^{k+m} e^{\sigma x} + x^k \tilde{Q}(x) e^{\sigma x} = \\ &= x^k (r_m x^m + \tilde{Q}(x)) e^{\sigma x} = x^k P(x) e^{\sigma x} \end{aligned}$$

ko'rinishdagi yechimga ega bo'ladi; bunda $P(x) = r_m x^m + \tilde{Q}(x)$, $\deg P(x) = m$. $\square$

Isbotlangan teoremlardan $L[y] = (b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0) e^{\sigma x}$ tenglamaning xususiy yechimini topish uchun quyidagi **noma'lum koeffitsientlar metodidan** foydalanish mumkinligi kelib chiqadi:

1⁰. $L(\sigma)$ ni hisoblaymiz. Agar $L(\sigma) \neq 0$ bo'lsa, $k=0$ deymiz, aks holda esa, ya'ni $L(\sigma) = 0$ bo'lganda k bilan σ ildizning karralilik darajasini belgilaymiz.

2⁰. Tenglamaning yechimini $y = x^k (r_m x^m + r_{m-1} x^{m-1} + \dots + r_1 x + r_0) e^{\sigma x}$ ko'rinishda izlaymiz; bunda $r_m, r_{m-1}, \dots, r_1, r_0$ - hozircha noma'lum koeffitsientlar.

3⁰. Bu y ni berilgan tenglamaga qo'yib, chap tomonni soddalashtirib, tenglamani $e^{\sigma x}$ ga qisqartirib, ko'phadlarning tengligini hosil qilamiz.

4⁰. Chap va o'ng tomondagi kophadlardagi x ning bir xil darajalari oldidagi koeffitsientlarni tenglashtirib, $r_m, r_{m-1}, \dots, r_1, r_0$ - noma'lum koeffitsientlarga nisbatan chiziqli algebraik tenglamalar sistemasini topamiz.

5⁰. Bu sistemadan noma'lum koeffitsientlarni aniqlaymiz (sistemaning yechimga egaligini teoremlar ta'minlaydi).

6⁰. Koeffitsientlarning topilgan qiymatlarini $y = x^k(r_m x^m + r_{m-1} x^{m-1} + \dots + r_1 x + r_0) e^{\sigma x}$ ga qo'yib, izlangan xususiy yechimni hosil qilamiz.

Misol 4. Tenglamani yeching

$$y'' - (1+2i)y' + (-1+i)y = 2xe^{ix}.$$

□ Berilgan tenglamaning umumiy yechimi mos bir jinsli tenglamaning umumiy yechimiga berilgan bir jinsli bo'lmagan tenglamaning xususiy yechimini qo'shishdan hosil bo'ladi. Bir jinsli tenglamaning xarakteristik ko'phadi: $L(\lambda) = \lambda^2 - (1+2i)\lambda - 1+i$. Uning ildizlari $\lambda_1 = i, \lambda_2 = 1+i$. Bir jinsli tenglamaning umumiy yechimi

$$y = c_1 e^{ix} + c_2 e^{(1+i)x} \quad (\{c_1, c_2\} \subset \mathbf{C}).$$

Endi berilgan bir jinsli bo'lmagan tenglamaning xususiy yechimini topamiz. Tenglamaning o'ng tomoni $P(x)e^{\sigma x}$ uchun $\sigma = i, P(x) = 2x, \deg P(x) = 1$. Ravshanki i soni $L(\lambda) = 0$ tenglamaning $k = 1$ karrali (oddiy) ildizi. Demak, xususiy yechimni $y = x(r_1 x + r_0) e^{ix}$ ko'rinishda izlash mumkin. Bu y ni va uning $y' = (ir_1 x^2 + (2r_1 + ir_0)x + r_0) e^{ix}$, $y'' = (-r_1 x^2 + (-r_0 + i4r_1)x + 2r_1 + r_0) e^{ix}$ hosilalarini

berilgan tenglamaga qo'yib, e^{ix} ga qisqartirib, chap tomondagi o'xshash hadlarni ixchamlab, topamiz:

$$-2r_1 x + 2r_1 - r_0 = 2x.$$

Bundan $r_1 = -1, r_0 = 2r_1 = -2$ bo'lishi kelib chiqadi. Bu qiymatlarni $y = x(r_1 x + r_0) e^{ix}$ ga qo'yib, berilgan tenglamaning $y = -x(x+2) e^{ix}$ xususiy yechimini hosil qilamiz. Uning umumiy yechimi esa

$$y = c_1 e^{ix} + c_2 e^{(1+i)x} - x(x+2) e^{ix} \quad (\{c_1, c_2\} \subset \mathbf{C})$$

formula bilan beriladi.

Endi haqiqiy sohada berilgan va o'ng tomoni maxsus ko'rinishga ega bo'lgan ushbu

$$L[y] = e^{\alpha x} (P(x) \cos \beta x + Q(x) \sin \beta x) \quad (13)$$

tenglamaning xususiy yechimini topishda to'xtalamiz. Bunda $L[\circ]$ operatorning koeffitsientlari va α, β - haqiqiy sonlar; $P(x), Q(x)$ - haqiqiy koeffitsientli ko'phadlar va $\deg P(x) = m_1, \deg Q(x) = m_2$. Kompleks sohaga o'tib Eyler formulalariga ko'ra quyidagini yozamiz:

$$\begin{aligned} e^{\alpha x} (P(x) \cos \beta x + Q(x) \sin \beta x) &= e^{\alpha x} (P(x) \operatorname{Re} e^{i\beta x} + Q(x) \operatorname{Re}(-ie^{i\beta x})) = \\ &= \operatorname{Re}((P(x) - iQ(x))e^{(\alpha+i\beta)x}). \end{aligned}$$

Demak,

$$L[y] = (P(x) - iQ(x))e^{(\alpha+i\beta)x}$$

tenglamaning $y = x^k(r_m x^m + r_{m-1} x^{m-1} + \dots + r_1 x + r_0) e^{(\alpha+i\beta)x}$ (bunda $m = \max\{m_1, m_2\}$, agar $L(\alpha+i\beta) \neq 0$ bo'lsa, $k=0$, $L(\alpha+i\beta) = 0$ bo'lganda esa k soni $\sigma = \alpha+i\beta$

ildizning karralilik darajasi) ko'rinishdagi kompleks yechimini topib, unung haqiqiy qismini hisoblasak, berilgan (13) tenglamaning xususiy yechimini topgan bo'lamiz.

Misol 5. Ushbu

$$y'' - 4y' + 5y = e^x(3\cos x + \sin x) \quad (14)$$

tenglamaning xususiy yechimini toping.

□ Tenglamaning o'ng tomonini kompleks funksiyaning haqiqiy qismi ko'rinishida ifodalaymiz:

$$e^x(3\cos x + \sin x) = e^x(3\operatorname{Re} e^{ix} - \operatorname{Re}(ie^{ix})) = \operatorname{Re}((3-i)e^{(1+i)x})$$

Endi ushbu

$$y'' - 4y' + 5y = (3-i)e^{(1+i)x} \quad (15)$$

tenglamaning xususiy yechimini topamiz. Bu tenglamaga mos bir jinsli tenglamaning xarakteristik ko'phadi $L(\lambda) = \lambda^2 - 4\lambda + 5$ bo'lgani uchun $\sigma = 1+i$ uning ildizi emas.

Demak, (15) differensial tenglamaning xususiy yechimini $y = re^{(1+i)x}$ ko'rinishda izlash mumkin. Bu $y = re^{(1+i)x}$ ni tenglamaga qo'yib, $e^{(1+i)x}$ ga qisqartirib, topamiz:

$$(1+i)^2 r - 4(1+i)r + 5r = 3-i$$

Bundan $r = 1+i$. Demak, $y = (1+i)e^{(1+i)x}$ funksiya (15) tenglamaning xususiy yechimi. Endi oxirgi funksiyaning haqiqiy qismini topamiz:

$$\operatorname{Re}((1+i)e^{(1+i)x}) = e^x \operatorname{Re}((1+i)(\cos x + i\sin x)) = e^x(\cos x - \sin x)$$

Topilgan $y = e^x(\cos x - \sin x)$ haqiqiy funksiya berilgan (14) tenglamaning xususiy yechimidir.

Biz yuqorida haqiqiy sohada berilgan (14) tenglamaning xususiy yechimini kompleks sohaga o'tib topdik. Xususiy yechimni kompleks sohaga chiqmasdan, ya'ni haqiqiy sohada turib ham topish mumkin. Buning uchun (13) ning yechimini

$$y = x^k e^{\alpha x} (M(x)\cos \beta x + N(x)\sin \beta x) \quad (16)$$

ko'rinishda izlash kerak; bunda k - yuqorida aytilgan ma'noga ega, $M(x)$ va $N(x)$

ko'phadlarning darajalari esa $\leq m = \max\{m_1, m_2\}$. Ularning noma'lum koefitsientlarini topish uchun (16) funksiyaning (15) tenglamaga qo'yib, chap va o'ng tomondagi o'xshash hadlardagi koefitsientlarni tenglashtirib, hosil bo'lgan chiziqli algebraik tenglamalar sistemasini yechish kerak.

Misol 6. Ushbu

$$y'' - 4y' + 5y = 4e^{2x}(x\cos x + \sin x)$$

tenglamaning xususiy yechimini toping.

□ Ravshanki, $\sigma = 2+i$ son $\lambda^2 - 4\lambda + 5 = 0$ xarakteristik tenglamaning $k = 1$ karrali ildizi, $m_1 = 1, m_2 = 0, m = \max\{m_1, m_2\} = 1$. Demak, berilgan tenglamaning yechimini $y = xe^{2x}((ax+b)\cos x + (cx+d)\sin x)$ ko'rinishda izlash mumkin; bunda

a, b, c, d - hozircha noma'lum sonlar. Bu $y = e^{2x}((ax^2 + bx)\cos x + (cx^2 + dx)\sin x)$ ni

tenglamaga qo'yib, o'xshash hadlarni ixchamlab va $2e^{2x}$ ga qisqartirib, topamiz:

$$2cx\cos x - 2ax\sin x + (a+d)\cos x + (c-b)\sin x = 2x\cos x + 2\sin x$$

Bundan, chap va o'ng tomondagi o'xshash hadlardagi koeffitsientlarni tenglashtirib, $2c = 2$, $-2a = 0$, $a + d = 0$, $c - b = 2$ shartlarga ega bo'lamiz. Oxirgi sistemadan $a = 0$, $b = -1$, $c = 1$, $d = 0$ qiymatlarni topamiz. Bularni $y = xe^{2x}((ax + b) \cos x + (cx + d) \sin x)$ ga qo'yib, berilgan tenglamaning $y = xe^{2x}(x \sin x - \cos x)$ xususiy yechimini topamiz.

Mavzuni mustahkamlash uchun savollar:

1. Noma'lum koeffitsiyentlar metodi deb nimaga aytiladi?
2. Superpozitsiya prinsipi deb nimaga aytiladi?